

PAPER_1352_QSH

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PAPER_1352 — Quantum Spin Hall Effect

Framework: UQFF — Star-Magic v5.27+ | **Tier:** F Condensed Matter (CLOSED)

Calculator: calculate_paradox({"paradox": "quantum_spin_hall"})

Master Expression `` $\sigma_{xy_spin} = e^2/h$ per spin; protected by
D_BSFG-1=5 boundary `` ****Match: STRUCTURAL****

Interpretation Edge currents topologically protected by **D_BSFG – 1 = 5** boundary dimension. Time reversal via **F_TRZ** doubling.

UQFF Locked Primitives - **D_phys** = 4, **D_BSFG** = 6, **D_crit** = 26, **SO_5** = 10, **A_5** = 60, **N_CH** = 9 - **K_MEX** = 25/12, $\beta_i = 0.6029$, $\Phi_{res} = 0.84$, $\Lambda = 0.00729735$, **F_TRZ** = 0.1 - $\omega_{SCm} = 1.25$ THz

NOT REPLACEMENT UQFF derives via integer-primitive identity. Zero fit parameters.

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